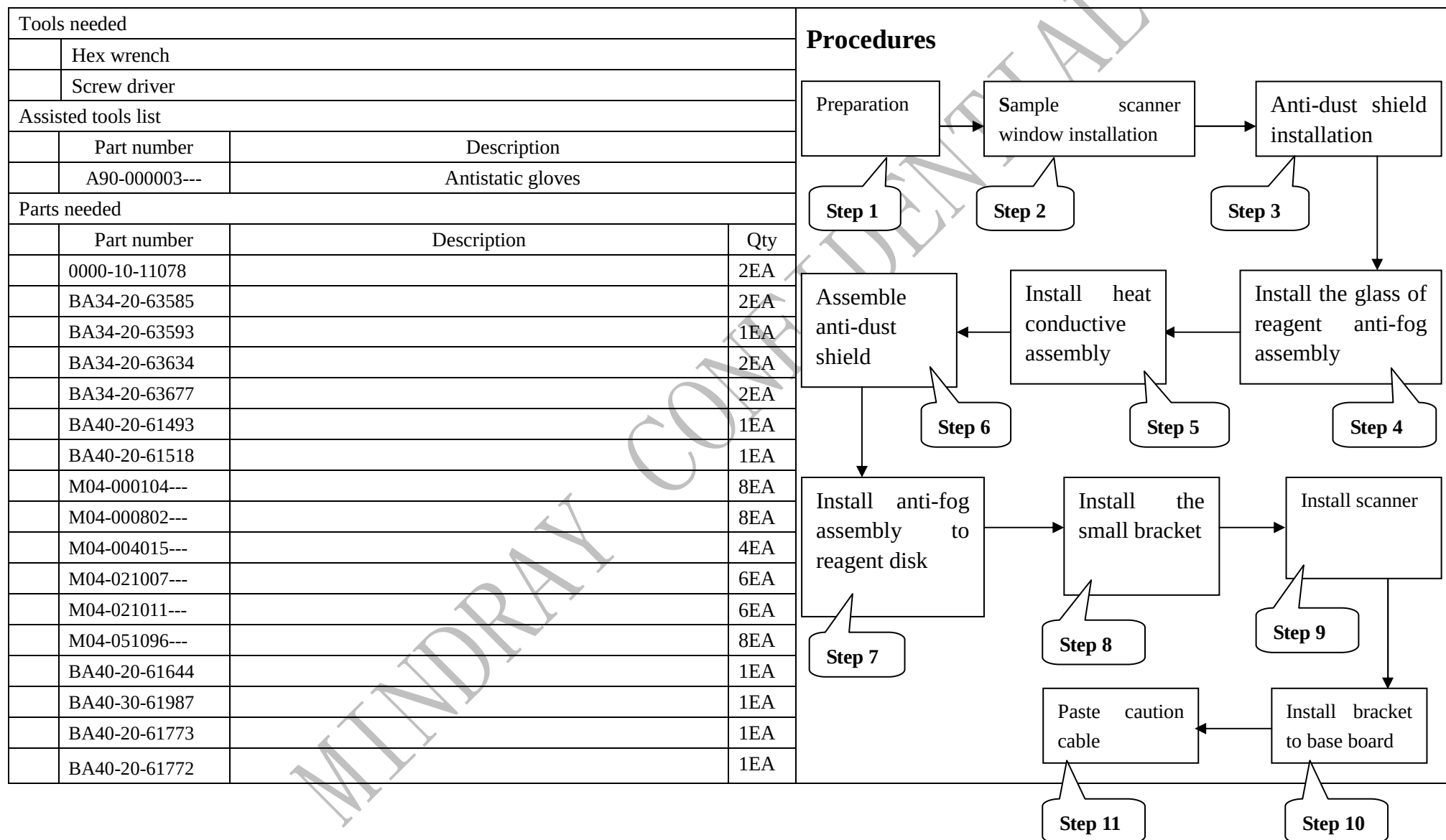


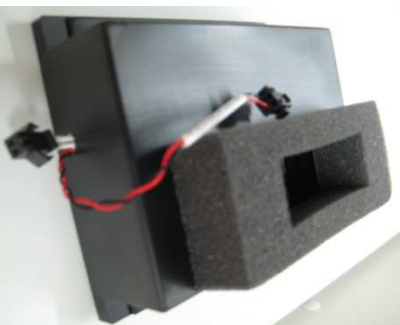
BS-480&BS-490 Internal Barcode Reader Installation Guide

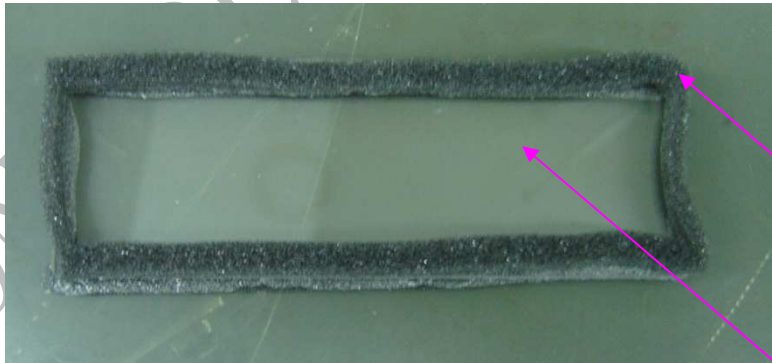
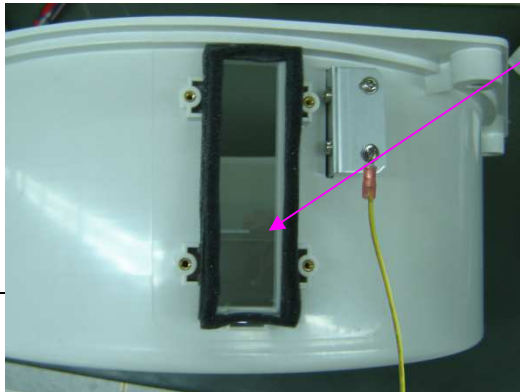
Version 1.0 Unicorn / Raul

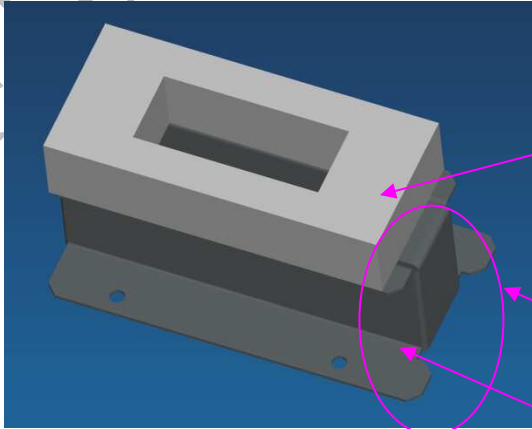
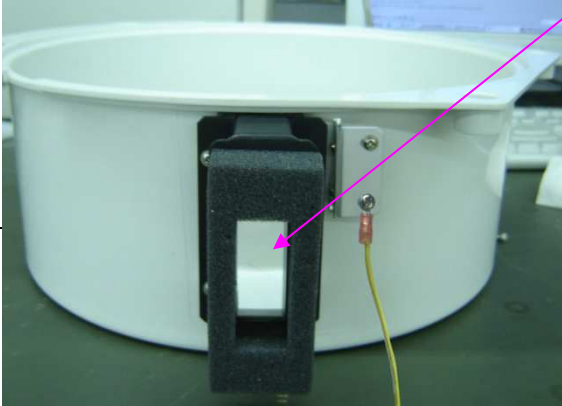
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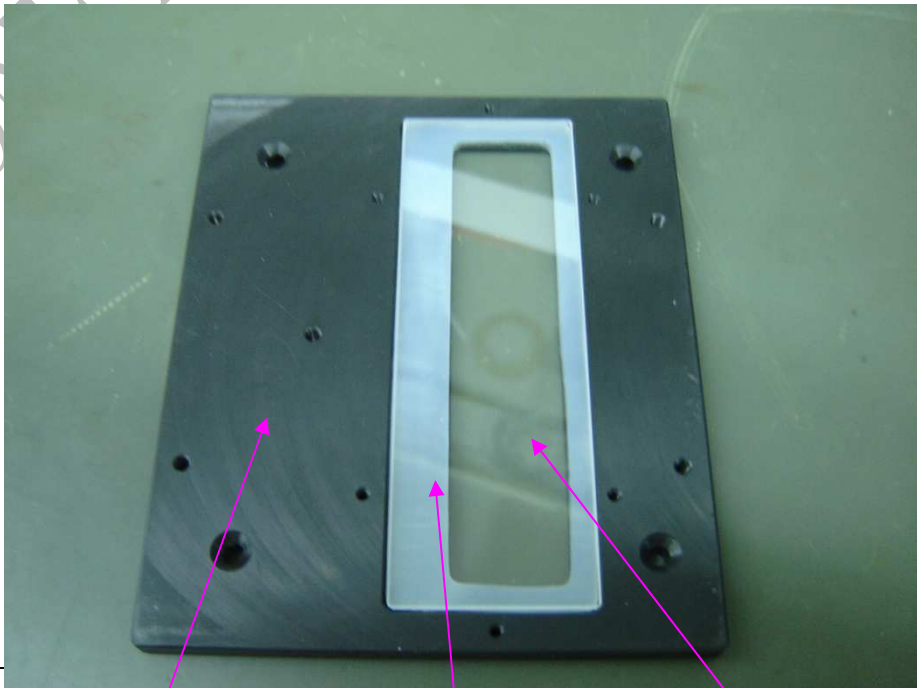


Step1: preparation			Schematic diagram	
Tools	Antistatic gloves		Pic1	
Procedure description	Get parts ready as shown in Pic1.		 Bar cord reader (1 pcs)	 Small bracket (1 pcs)
Note	1. Make sure the glass window is clean. 2. Ware antistatic gloves. It's not allowed to touch the window by finger directly.		 Reader bracket	 anti-fog assembly

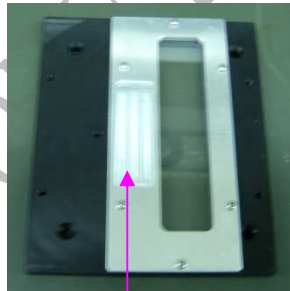
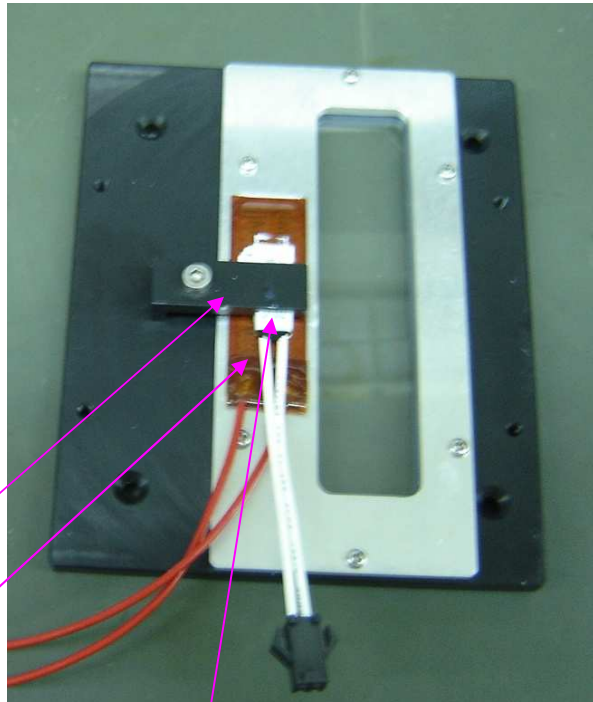
Check	1. Holder is bended or not; 2. Glass window is clean or not, if not, please use alcohol cotton to clean it.	
Step 2: Sample scanner window installation		Schematic diagram
Tools	Antistatic gloves	Pic2  sponge cushion2 glass Pic3 
Procedure description	1. Remove the scanner window cover on sample disk. 2. As shown in Pic2: stick sponge cushion2(BA34-20-63677) to the glass window(BA34-20-63593), make sure the cushion is aligned to the window edge. 3. As shown in Pic3: put the assembled glass to the sample compartment.	
Note	1. Keep the cover and screws, back up for future use. 2. Keep the window clean.	

Check	Glass is clean and there is no crack on it.	
Step 3: Anti-dust shield installation		Schematic diagram
Tools	Screw driver	Pic4  Sponge cushion1 This end is in the lower place when installed in the compartment Anti-dust shield Pic5 
Procedure description	- As shown in Pic4: stick the sponge cushion1(BA40-20-61518) in alignment to the anti-dust shield(BA40-20-61493), otherwise it may affect the light - As shown in Pic5: fix the anti-dust shield with 4 M3X8 screws to the sample compartment.	
Note	- The window of sponge cushion1 and the anti-dust shield must be aligned - make sure the position of the anti-dust shield is correct	

Check		
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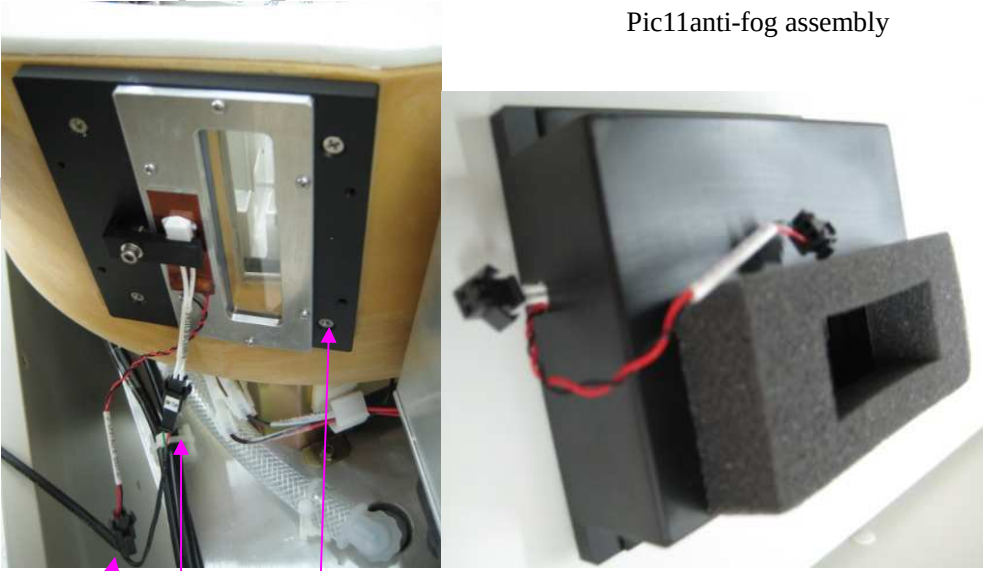
Step 4: install the glass of reagent anti-fog assembly		Schematic diagram
		Pic6
Procedure description	- As shown in Pic6, stick rubber cushion to the base panel, make sure the rubber is aligned to the panel - Put glass to the rubber cushion. Make sure the glass is clean and complete, otherwise it may affect the light	

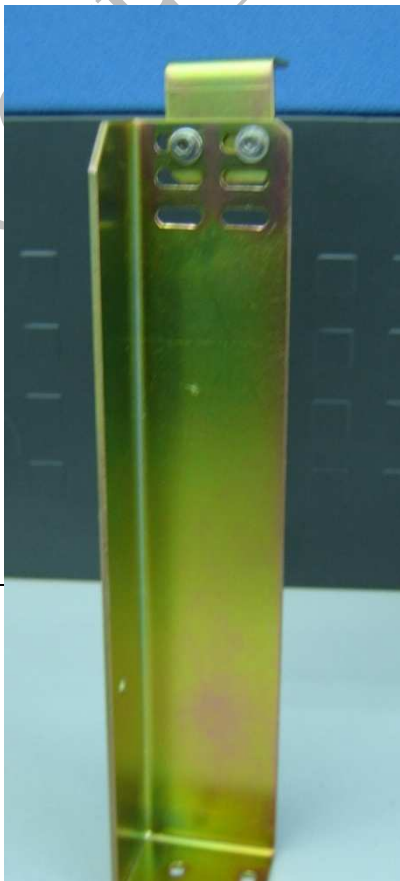
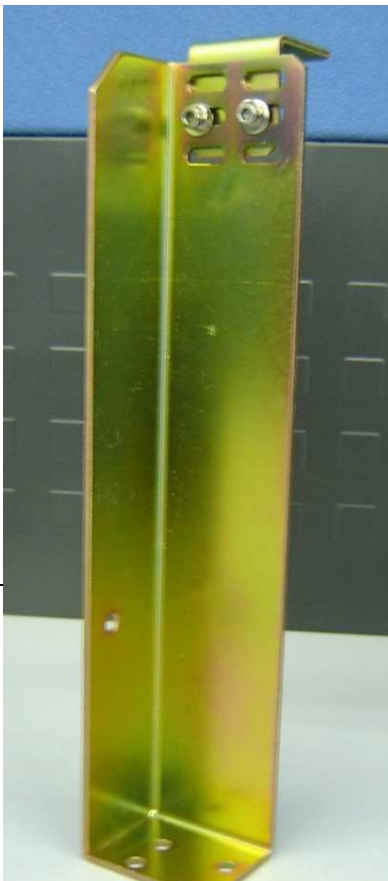
	Glass is clean and there is no crack on it.	Base panel	Glass window
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Step5: install heat conductive assembly					Schematic diagram	
tools	Screw driver		other tool	Thermal paste KP97	Pic7	Pic8
	Hex wrench					
Procedure description	1. As shown in pic7: fix the heat conductive aluminum plate to the panel. Do not use excessive force or the glass might be broken 2. As shown in pic8: 1) Add a heater to the panel, the direction of the cable connected to the heater and heat protective switch should be the same with the picture. 2) Add thermal paste to the heat protective switch before it is adhered to the heater. Use screw to fix the pressing plate, do not use excessive force or it may damage the switch				 aluminum plate	 pressing plate

check	1. The direction of the cable is correct	heater Heat protective switch
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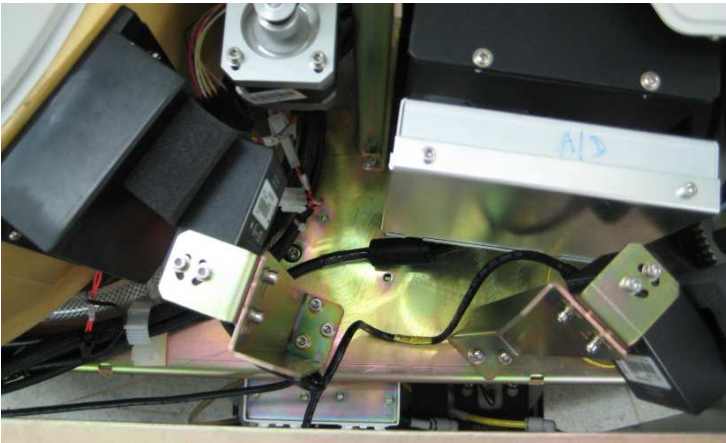
Step 6: assemble anti-dust shield					Schematic diagram
					Pic9 
Procedure description	As shown in pic9: 1. Adhere sponge to the anti-dust shield, make sure the sponge window is aligned to the shield window, otherwise it may affect the light				

check	1. make sure the sponge window is aligned to the shield window	
Step 7: install anti-fog assembly to reagent disk		Schematic diagram
		Pic10
Procedure description	- As showed in Pic10, fix the base panel on reagent disk with 4 pieces of M3X10 screws - Connect cable (BA40-20-61644) to J15 of reagent refrigeration board. - And then fix the anti-fog assembly (as Pic11) to reagent disk, finally as Pic12.	Pic11 anti-fog assembly  Pic12
Note		

Check		Base pannel Over heat protector Heater 
Step 8: install the small bracket		Schematic diagram
Tools		 
Procedure description	As Pic13, fix the small bracket onto the big bracket with screw (M04-051096---), with shim(M04-000802---) and spring shim(M04-000104---).	

Note	1. Two small brackets, one for sample, other one for reagent. 2. NOT tighten the screws before scanning function test and alignment adjusting.	
Check		
Step 9: install scanner		Schematic diagram
Tools		Pic14
Procedure description	As Pic14, fix the scanner onto the small bracket with screw (M04-051096---), with shim (M04-000802---) and spring shim (M04-000104---).	

Note	1. Keep the window of scanner clean, not touch by finger. 2. NOT tighten the screws before scanning function test and alignment adjusting.	
Check	Glass window is clean or not, if not, please use alcohol cotton to clean it.	
Step 10: install bracket to base board		Schematic diagram
Tools		Pic15

Procedure description	1. As Pic15, fix the bracket onto the base board with screw (M04-051062---), with shim (M04-021011---) and spring shim(M04-021007---). 2. Connect sample barcode cable BA38-20-88168 to J11 of Probe drive board, connect reagent barcode cable BA38-20-88169 to J12 of Probe drive board	
Note		
Check		

Step 11: paste caution cable	Schematic diagram
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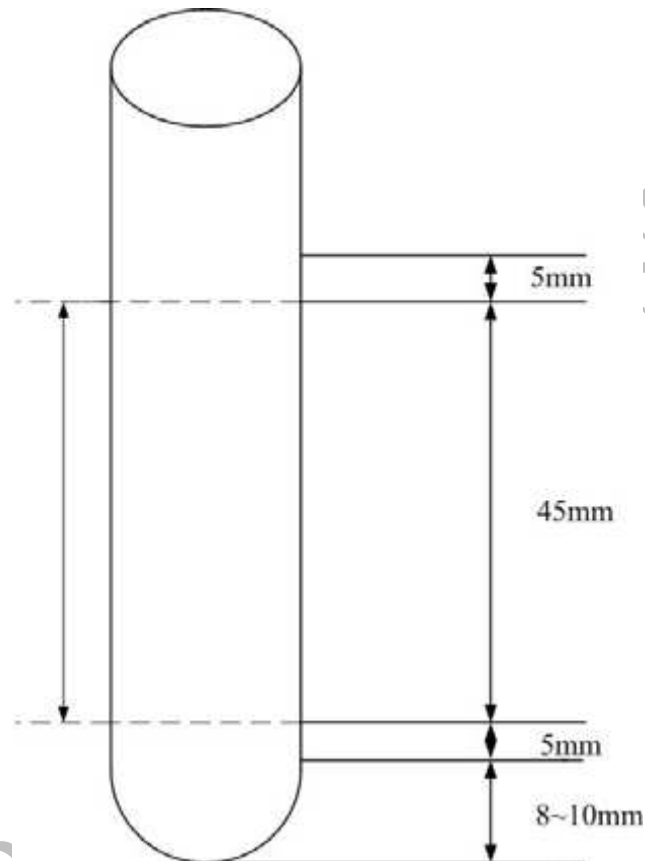
Tools		Pic16  paste caution cable
Procedure description	As Pic16 showed, the CAUTION label (BA40-20-73064) should be pasted around scanning window.	
Note	CAUTION cable should be pasted near scanning window.	
Check		

Sample Barcode Scanning Function Adjustment

Preparation

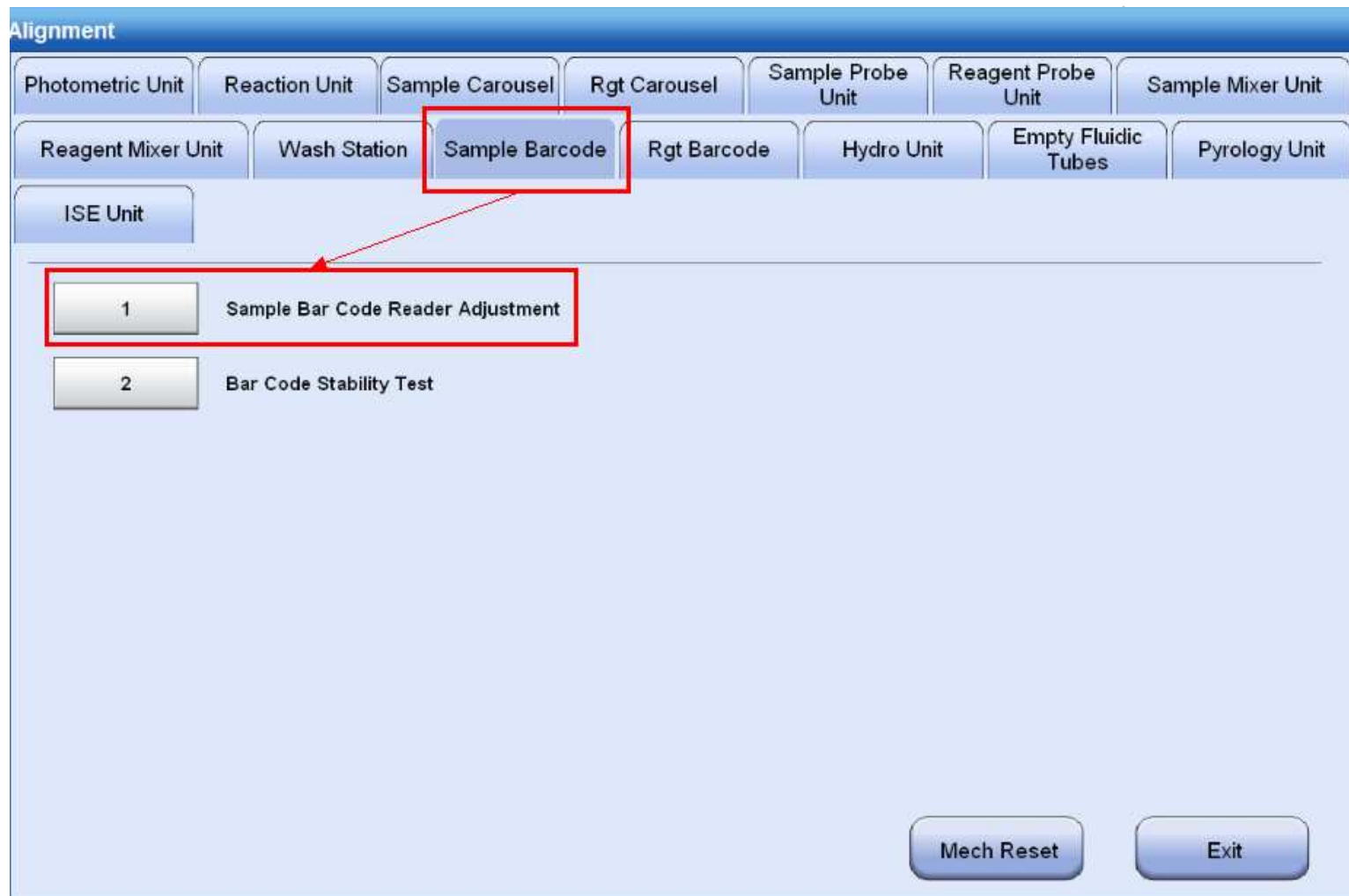
Check cable connection is OK; Make sure glass window is clean. If not, clean it with alcohol cotton. Make sure the foam does not block the glass window.

Note: the barcode quality level should above C, barcode is clear and not stained. Barcode is pasted according to below picture.



Barcode reader scanning position adjustment

- 1) Choose “Utilities→Maintenance→Alignment→ Sample Barcode→ Sample Bar Code Reader Adjustment”.



- 2) Place Bar Code alignment tool in position #31 of sample carousel middle ring.

Alignment Procedure

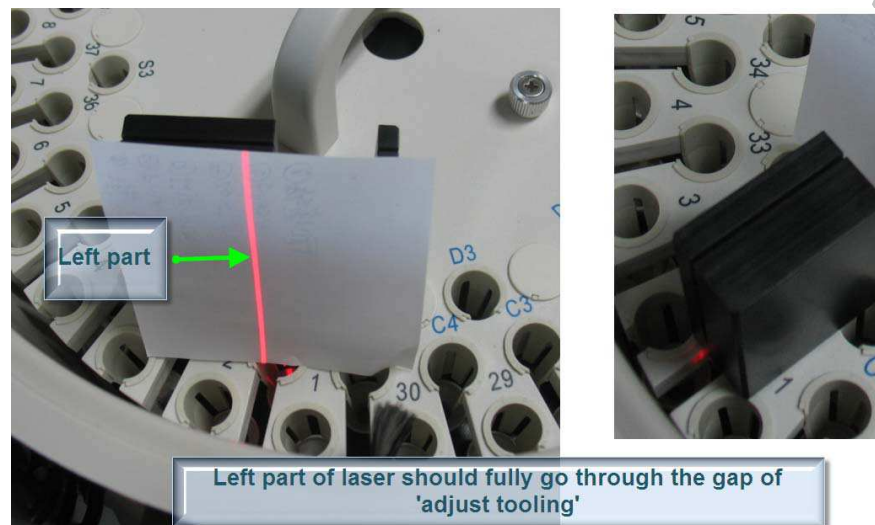
Current Step **1**

1.	Prerequisites: None. Postrequisites: None. Place bar code alignment tool in position #31 of sample carousel middle ring.
2.	Check the light position. If light does not go through the fixture slot, loosen the two screws retaining the barcode reader to adjust incident angle. Adjust the sample carousel with the left and right arrow buttons. If light beam is wider than the fixture slot, adjust the installation position of the barcode reader, allowing the left of the light beam (counter-clockwise) to go through the slot. If light beam is not vertical, adjust the two screws retaining the barcode reader's small bracket.
3.	Make sure the scanning light goes through the fixture slot. Otherwise, adjust with on screen left/right arrow buttons till requests are met.
4.	Remove alignment tool from position #31, load bar-coded sample cups in position #1-#6 and #31-#35, and then check if the sample bar code is correct during scanning.
5.	Check if the returned bar code is the same as the actual one. If not, check if the light position is correct and the bar code reader works normally.

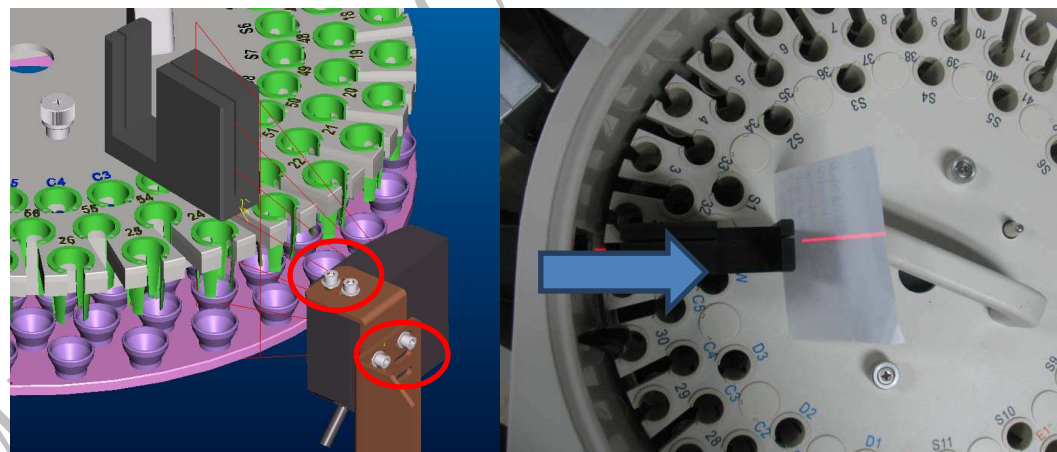
Continue **Stopped**

3) If the light is not vertical, please adjust the screws on small bracket of bar code reader. If the width of light is wider than the gap of

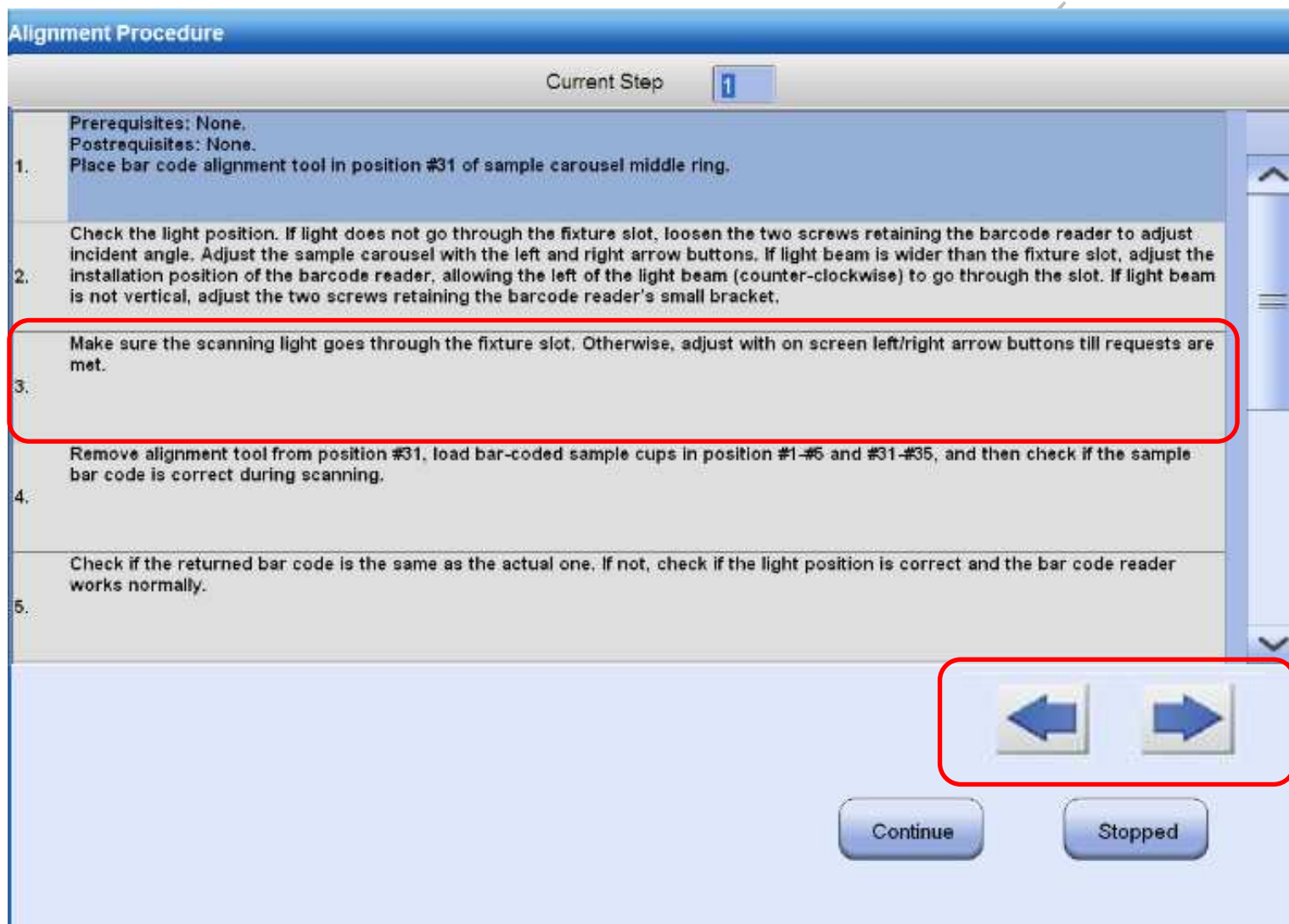
adjustment tool, please adjust the position of bar code reader to make sure the LEFT part of light go through the gap of adjustment tool.



Picture below shows the correct position.



- 4) Make sure the scanning light goes through the fixture slot. Otherwise, adjust with the screen left/right arrow buttons till requests are met.



- 5) Remove alignment tool from position 31#, load bar-code sample cups in position 1-5# and 31-35#, and then check if the sample bar code is correct

during scanning. If not, check the light position and the sample bar code.

- 6) Click “Continue”, save the parameter and exit.

Sample Bar Code Stability Test

- 1) Load bar-code sample cups in position 1-60#, “Scan circle” fill 10, click Start to scan sample bar code.

Sample Bar Code Stability Test

	1	2	3	4	5
1#					
2#					
3#					
4#					
5#					
6#					
7#					
8#					
9#					
10#					
11#					
12#					

Scan Circle

Start Stop Exit

If the stability testing fails, please check as follows:

- - check the failure code is fit for the print standard or not; code is clear or not; code is pasted on right position of tube or not.
- - check the glass window is clean or not.
- - If everything is OK, please change the barcode reader and test again.

Reagent Barcode Scanning Function Adjustment

Preparation

Check serial cable connection is OK; Make sure glass window is clean. If not, clean it with alcohol cotton. Make sure the foam does not block the glass window; Ensure the defog function is working (Touch the fixing board from outside, if it's warm, it's ok); Check the sponge is not deformed and not block the window of scanner.



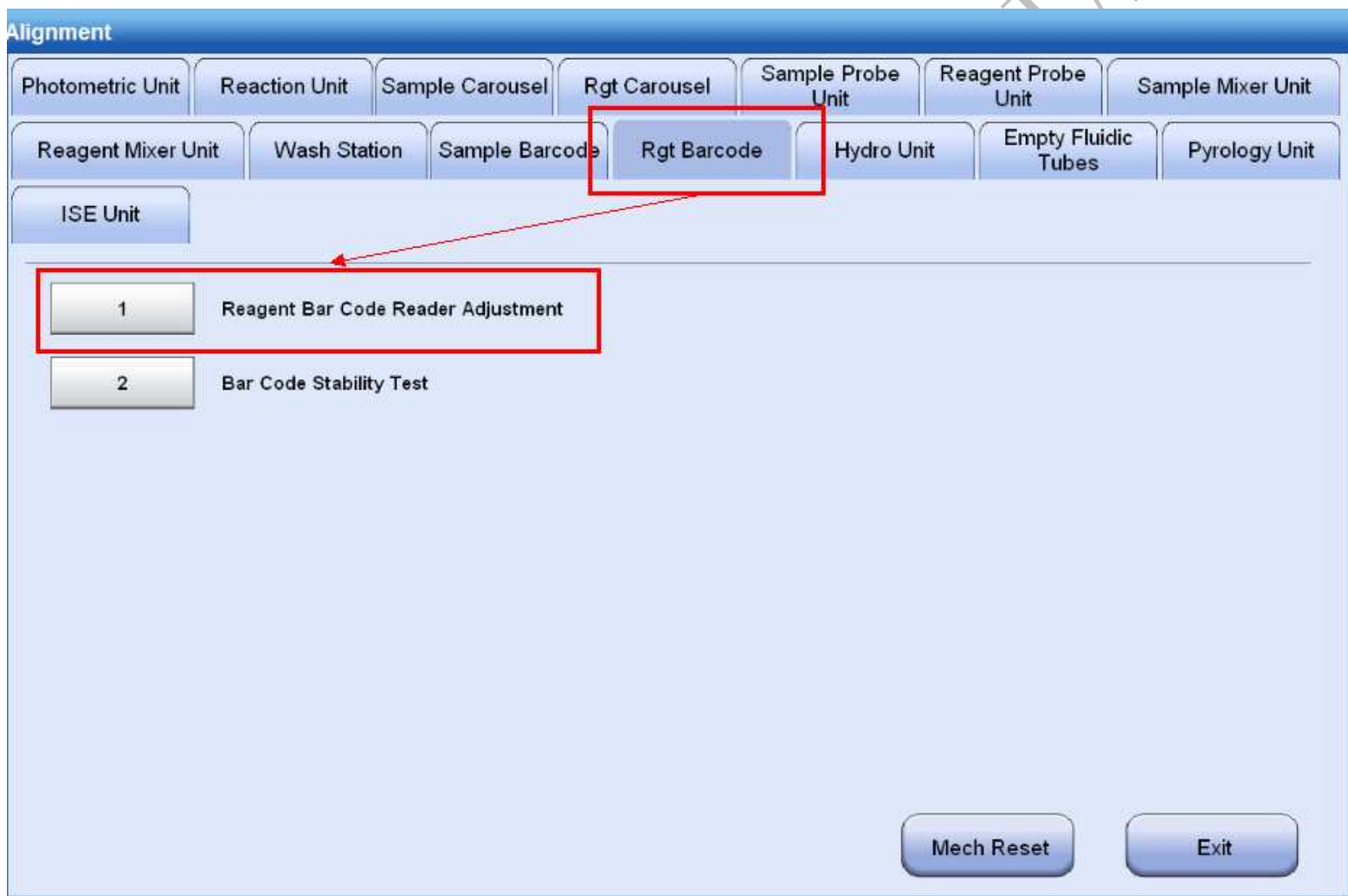
Reagent adjustment tool #1



Reagent adjustment tool #2

Barcode reader scanning position adjustment

- 1) Choose “Utilities→Maintenance→Alignment→ Reagent Barcode→ Reagent Bar Code Reader Adjustment”.



- 2) Make sure **Reagent adjustment tool #1** can put in the gap between Reagent carousel #1 and #2 easily, if not, adjust Reagent carousel position with the screen left/right arrow buttons.

Alignment Procedure

Current Step **1**

Prerequisites: None.
Postrequisites: None.

- Place bar code alignment tool in position #43 of reagent carousel inner ring.
- Check the light position. If light does not go through the fixture slot, loosen the two screws retaining the barcode reader to adjust incident angle. If light beam is wider than the fixture slot, adjust the installation position of the barcode reader, allowing the left of the light beam (counter-clockwise) to go through the slot. If light beam is not vertical, adjust the two screws retaining the barcode reader's small bracket.
- Make sure the scanning light goes through the fixture slot. Otherwise, adjust with on screen left/right arrow buttons till requests are met.

Remove alignment tool from position #43 and load bar-coded reagent bottles to position #1-#5 and #43-#47.

-
-
-
-
-

Check if the light position is correct and the bar code reader

Reagent adjustment tool #1 →

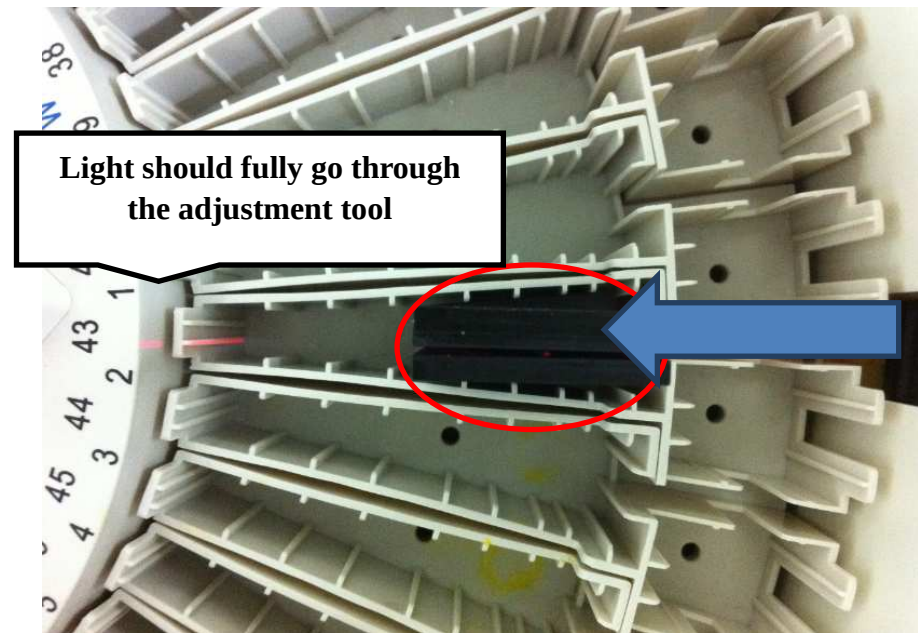
#2

#1

← →

Continue Stopped

3) Place **Reagent alignment tool #2** in position #43 of Reagent carousel. Make sure the scanning light go through the fixture slot. Otherwise, loosen the two screws retaining the barcode reader to adjust incident angle.





- 4) Make sure the scanning light go through the fixture slot. Otherwise, adjust with the screen left/right arrow buttons till requests are met.

Alignment Procedure

Current Step 1

1. Prerequisites: None.
Postrequisites: None.
Place bar code alignment tool in position #43 of reagent carousel inner ring.
2. Check the light position. If light does not go through the fixture slot, loosen the two screws retaining the barcode reader to adjust incident angle. If light beam is wider than the fixture slot, adjust the installation position of the barcode reader, allowing the left of the light beam (counter-clockwise) to go through the slot. If light beam is not vertical, adjust the two screws retaining the barcode reader's small bracket.
3. Make sure the scanning light goes through the fixture slot. Otherwise, adjust with on screen left/right arrow buttons till requests are met.
4. Remove alignment tool from position #43 and load bar-coded reagent bottles to position #1-#5 and #43-#47.
5. Check if the returned bar code is the same as the actual one. If not, check if the light position is correct and the bar code reader works normally.

← →

Continue Stopped

- 5) Remove **Reagent alignment tool #2** from position #43 and load barcode reagent bottles to position #1-#5 and #43-#47. and then check if the Reagent bar code is correct during scanning.If not, check the light position and the reagent bar code.

Alignment Procedure

Current Step: 1

Prerequisites: None.
Postrequisites: None.

1. Place bar code alignment tool in position #43 of reagent carousel inner ring.
2. Check the light position. If light does not go through the fixture slot, loosen the two screws retaining the barcode reader to adjust incident angle. If light beam is wider than the fixture slot, adjust the installation position of the barcode reader, allowing the left of the light beam (counter-clockwise) to go through the slot. If light beam is not vertical, adjust the two screws retaining the barcode reader's small bracket.
3. Make sure the scanning light goes through the fixture slot. Otherwise, adjust with on screen left/right arrow buttons till requests are met.
4. Remove alignment tool from position #43 and load bar-coded reagent bottles to position #1-#5 and #43-#47.
5. Check if the returned bar code is the same as the actual one. If not, check if the light position is correct and the bar code reader works normally.

Continue Stopped

- 6) Click "Continue", save the parameter and exit.